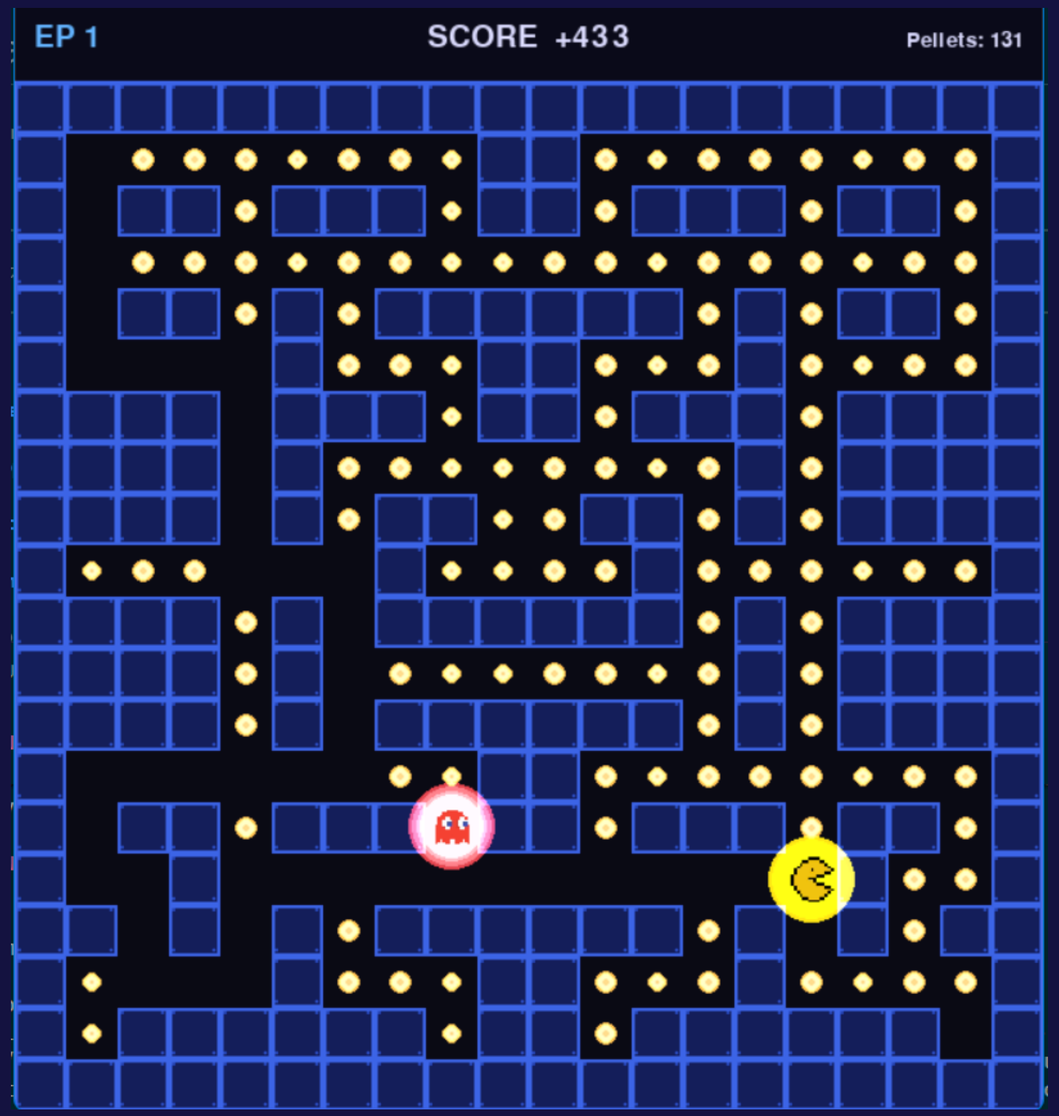
LET'S PLAY
PACMAN
START

PACMAN USING DEEP REINFORCEMENT LEARNING



OBJECTIVES

Learn an optimal decision-making policy under risk
Train Pacman to balance two competing objectives —collecting pellets (reward) while avoiding the ghost (risk), rather than just maximizing short-term gains.

Develop long-term planning and spatial awareness
Enable the agent to move beyond immediate actions and learn route-level strategies, safe navigation, and corridor-based planning across the maze.

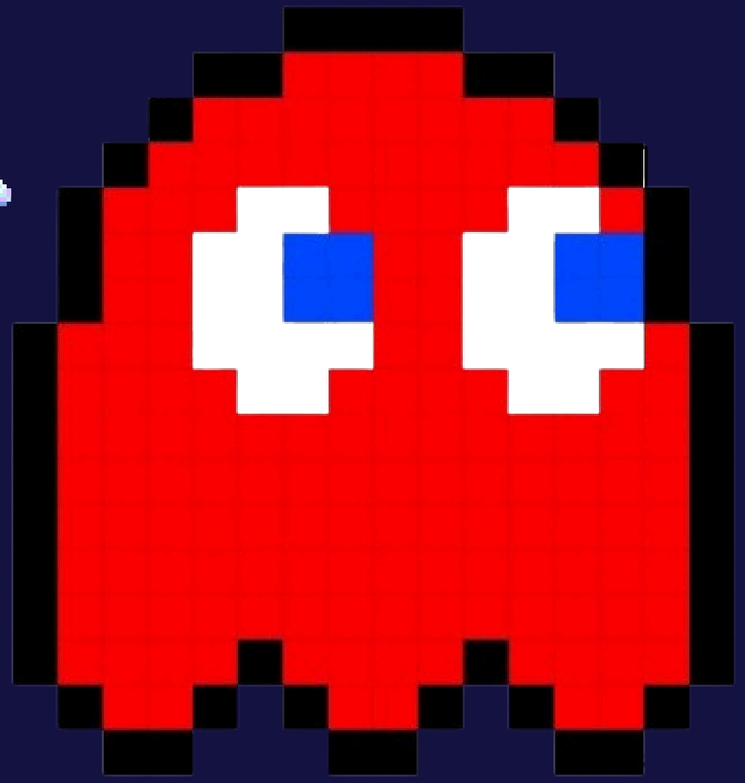
Build a stable and practical deep reinforcement learning system
Implement a complete DQN pipeline (with replay buffer, target network, curriculum learning, etc.) that can train reliably in a non-trivial environment.

SURVIVAL VS REWARD

GAME RULES

- THE GAME IS PLAYED ON A 20X20 MAZE CONTAINING WALLS AND PELLETS. PACMAN'S GOAL IS TO COLLECT ALL PELLETS WHILE AVOIDING THE GHOST.
- THE GHOST CAN MOVE IN 4 DIRECTIONS. UP, DOWN, LEFT, RIGHT.
- RULE-BASED GHOST (USES A* SEARCH ALGORITHM AT ITS CORE). HALF THE SPEED OF PACMAN
- IF PACMAN COLLECTS ALL PELLETS, PACMAN WINS. IF IT GETS CAUGHT BEFORE THEN GHOST WINS.
- COLLECTING PELLETS GIVES POSITIVE REWARD. GETTING CAUGHT GIVES A LARGE NEGETIVE PENALTY. SMALL PENALTIES FOR BAD DECISIONS.

WHAT I'VE DONE



- Designed a custom RL environment
- Built a 20×20 grid-based Pacman environment with pellets, walls, and a deterministic ghost, turning the problem into a structured sequential decision-making task.
- Developed a convolutional Deep Q-Network (DQN)
- Implemented a CNN that takes a 5-channel state (walls, pellets, Pacman, ghost, timing cue) and extracts spatial features like corridors, danger zones, and escape paths.
- Used a dueling network architecture
- Split the network into value V(s)V(s)V(s) and advantage A(s,a)A(s,a)A(s,a) streams to compute Q-values, improving stability and helping the model distinguish between good states and good actions.
- Epsilon-greedy exploration, curriculum learning, randomised starts during training so the ghost learns multiple patterns.

THANK YOU